

# Response to the Editor and Reviewers

## An Accuracy and Calibration Benchmark of scGPT Pipelines and Matched Classical Models for Cell Type Annotation

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Dear Dr Jadhav and Reviewers,

Thank you for the careful and constructive assessment. We addressed the reports through new computation and a complete publication-facing rewrite. The central evaluation is now donor-held-out in every human dataset, hyperparameters are selected independently for each representation inside the training donors, and the analysis distinguishes the effect of the representation from that of the downstream classifier. The manuscript contains no language that assumes access to an earlier version; it reads as a standalone scientific paper.

The principal changes are:

1. a 16-word punctuation-free scientific title;
2. five-fold donor-held-out evaluation in three human atlases containing 965,427 analysed cells and 312 donor identifiers;
3. official frozen scGPT 0.2.5 embeddings from a pinned source commit and SHA-256-pinned whole-human checkpoint;
4. training-only feature selection and identical vocabulary-matched genes for scGPT, normalized expression and SVD-512 representations;
5. independent training-only hyperparameter selection for all evaluated representation-classifier combinations;
6. primary classifier-matched comparisons rather than choosing the best head from test performance;
7. a 512-component expression control to distinguish representation-specific effects from generic dimensional compression;
8. 5,000-draw paired donor-cluster bootstrap comparisons, with their conditional interpretation stated explicitly;
9. class-specific metrics and confusion matrices, including focused biological analyses of lymphocyte-boundary and rare brain-cell errors;
10. labelled-only feature selection in the PBMC scarcity experiment;
11. held-out-donor temperature scaling for both Logistic Regression and XGBoost in every representation in TNBC and brain; and
12. separation of pig pancreas as a human-checkpoint cross-species stress test without orthology mapping.

The revised conclusion is deliberately pipeline-specific: matched expression is compared with fixed frozen-scGPT pipelines in within-atlas unseen-donor settings. The paper does not claim broader superiority over official scGPT fine-tuning, newer checkpoints, other foundation models, zero-shot discovery or external-atlas transfer.

## Editor comments

*Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained.*

**Response.** Every reported result was regenerated from raw counts with the official scGPT embedding interface and a fully executable workflow. The clean manuscript reports three complete human donor-held-out analyses as primary. It no longer bases the central conclusion on random cell splits. Feature selection, vocabulary matching, SVD fitting, parameter selection and classifier fitting are restricted to outer-training donors. The per-cell probability vectors and class order are retained for exact aggregation and audit.

Claims have been rewritten around the evaluated estimand. The Abstract, Introduction, Discussion and Conclusions identify the tested object as a complete frozen-embedding pipeline and distinguish unseen-donor prediction within a curated atlas from external validation. The limitations section addresses the fixed checkpoint, absence of official fine-tuning, broad portal labels, possible checkpoint pretraining overlap, absence of external-atlas transfer and conditional bootstrap inference.

**Changes in the manuscript.** These changes appear throughout the clean manuscript, particularly in “Study design and estimands”, “Outer donor-held-out partitions”, “Group-cluster bootstrap comparisons”, the complete Discussion and the Conclusions.

*The title should have a clear precise scientific meaning and should not contain punctuation and it should not exceed 20 words.*

**Response.** The title is now:

### **An Accuracy and Calibration Benchmark of scGPT Pipelines and Matched Classical Models for Cell Type Annotation**

It has 16 words, contains no punctuation, describes the evaluated pipelines and does not announce the outcome.

## Reviewer 1

We thank Reviewer 1 for the positive assessment and the two focused statistical recommendations. We implemented a stronger extension of both requests by using donor-held-out folds and donor-cluster uncertainty in three human atlases rather than in only one representative cohort.

### **Comment 1: dependence among cells**

*Bootstrap confidence intervals are obtained by resampling individual cells. Because cells originating from the same donor or biological sample are correlated this approach may underestimate uncertainty. It would therefore be helpful to repeat the bootstrap at the donor or sample level for one representative analysis.*

**Response.** The primary uncertainty analysis now uses the donor as the resampling unit in every human dataset. After generating one outer-fold prediction for each retained cell, we construct

one confusion matrix per donor. In each of 5,000 replicates, donors are sampled with replacement and all cells belonging to a selected donor move together. The same sampled donor multiplicities are used for both members of a paired representation comparison.

This design avoids treating thousands of correlated cells as thousands of independent biological replicates. It also permits direct intervals for the difference between matched expression and scGPT embeddings at each fixed classifier. The main manuscript does not use separate marginal intervals to infer a paired difference.

The limitation has not been hidden: the group bootstrap resamples fixed out-of-fold predictions and does not repeat feature selection, SVD, parameter selection or model fitting inside each draw. We therefore describe its interval as conditional on the completed five-fold fits.

**Changes in the manuscript.** The procedure and interpretation are in “Group-cluster bootstrap comparisons”, the primary Results table, the uncertainty paragraph of the Results and the Discussion. Full paired comparisons are supplied in Supplementary Information and CSV form.

## Comment 2: variability across train–test splits

*I recommend repeating one representative analysis across approximately five independently generated train–test splits. If feasible it would be particularly informative to perform these follow-up splits at the donor or sample level.*

**Response.** We implemented five-fold stratified group cross-validation by donor for three human atlases. The datasets contain 101, 199 and 12 unique donor IDs. Each donor is held out exactly once, no donor appears in both training and test data within a fold, and every class is required to occur in both partitions. The five folds are correctly described as cross-validation folds whose training sets overlap, not as five independent training samples.

Every data-dependent operation is repeated inside the outer fold. Fold-specific results and mean  $\pm$  standard deviation across folds are reported, while the pooled out-of-fold metric is separately identified. This distinction also explains why a fold mean and a pooled Macro F1 can differ.

**Changes in the manuscript.** The design is presented in Figure 1, Table 1, “Outer donor-held-out partitions”, the first Results subsection and Supplementary fold tables.

## Reviewer 2

We thank Reviewer 2 for identifying the central generalization, comparator and calibration issues. These comments motivated the donor-held-out primary design, independently tuned representations, SVD control, strict scarcity sensitivity and expanded calibration analysis.

## Major concern 1: realistic cross-sample annotation

*The datasets are split randomly at the cell level rather than by donor sample or batch. Consequently cells from the same biological sample may appear in both training and test sets introducing information leakage and producing an overly optimistic estimate of generalization.*

**Response.** Random cell splits are no longer the primary human analysis. Three human atlases use five donor-held-out folds, covering 965,427 analysed cells and 312 donor identifiers. Training-

only feature selection prevents held-out donors from influencing the gene list. Programmatic assertions reject donor overlap, class-incomplete folds and misaligned out-of-fold predictions.

Only pipelines with all five aligned outer prediction folds were admitted to the final aggregation. The earlier TME and human-pancreas random-split results were removed, and partial donor-fold outputs are not reported. The resulting primary evidence is smaller in atlas count but stronger in biological partitioning and auditability.

We describe the resulting estimand precisely: generalization to unseen donors within the same curated atlas. Donor separation does not remove shared study processing, annotation conventions or cohort composition, so the manuscript does not call this external or out-of-cohort validation.

**Changes in the manuscript.** The human random-split claim has been replaced by the donor-held-out factorial benchmark throughout the Abstract, Methods, Results and Discussion.

## Major concern 2: label-rich broad classes

*The benchmark setting may not represent the scenarios in which foundation models are expected to provide the greatest value. Evaluation on label-scarce settings rare or fine-grained populations and external-atlas transfer would be needed to support broader conclusions.*

**Response.** We added a strict label-scarcity sensitivity in Indonesia PBMC using the same outer donor folds. Within each outer-training partition, exactly 10, 50, 100, 250 or 500 labelled cells are sampled per class with two seeds. Gene filtering, HVG selection and vocabulary matching are recomputed from those labelled cells alone. Unlabelled outer-training cells do not contribute to representation construction or classifier fitting. Logistic Regression is tuned independently for scGPT embeddings and matched expression using only the labelled subset.

The manuscript describes this as supervised few-label adaptation to known broad categories. It does not substitute this experiment for rare-cell discovery, unseen classes or external-atlas transfer. Those regimes remain explicit open questions, and broad conclusions about foundation models have been removed.

**Changes in the manuscript.** The experiment is reported in the Methods subsection “Strict labelled-cell scarcity sensitivity”, the dedicated Results subsection, the Discussion and Supplementary Information.

## Major concern 3: calibration and the downstream head

*The probabilities used to calculate calibration are generated by classifiers fitted on scGPT embeddings rather than by scGPT itself. The conclusions should therefore be framed as applying to specific scGPT-based pipelines rather than to scGPT intrinsically. Post-hoc calibration and additional complementary calibration analyses would further strengthen this assessment.*

**Response.** We agree. Every accuracy and calibration result is attached to a complete representation–classifier pipeline. “scGPT” alone is reserved for the model, checkpoint, software or embedding operation. Logistic Regression, Random Forest and XGBoost are paired with all three representations, making the effect of the head visible rather than implicit.

Equal-width top-label ECE is now accompanied by adaptive ECE, classwise ECE, multiclass Brier score and NLL. Reliability tables retain the number of cells in every bin. We avoid treating one binned ECE as a complete assessment of probability quality.

We also added post-hoc temperature scaling for Logistic Regression and XGBoost in scGPT embeddings, matched expression and SVD-512 in TNBC and brain. Parameters are selected on an inner fitting and validation partition, a new classifier is fitted without the inner calibration donors, and the scalar temperature is fitted only on those held-out calibration donors before evaluation on the untouched outer-test donors. Assertions confirm that the predicted class does not change. Because these calibration models omit the calibration donors from classifier fitting, their discrimination rows are distinguished from the final models fitted on all outer-training donors. The manuscript explicitly presents this as a two-atlas sensitivity rather than a universal calibration result.

The earlier fine-tuning result used a random within-object split and could not be placed on the same donor-held-out, training-nested footing without a separate large fine-tuning study. We therefore removed it from the comparative evidence rather than mixing incompatible estimands. The clean manuscript states explicitly that official end-to-end scGPT fine-tuning remains outside the scope, so a result for the frozen embedding interface is not used to make a claim about fine-tuned scGPT.

**Changes in the manuscript.** Pipeline-specific language is used in the title, Abstract, figures, tables, Results, Discussion and Conclusions. Temperature-scaling and complementary-metric results are in the main calibration figure and Supplementary Information.

### Minor concern 1: promote the matched benchmark

*Figure 1 and Table 2 emphasize the initial less controlled comparison whereas the vocabulary- and learner-matched analysis is the more appropriate primary benchmark.*

**Response.** Done. The primary figure and tables now show nine fixed scGPT-embedding versus matched-expression comparisons: three classifiers in each of three atlases. Every representation uses the same training-selected genes, and comparisons are made at fixed classifier. SVD-512 is a secondary control with Logistic Regression in all three atlases and with all classifiers in TNBC and brain. An unmatched analysis is not used to support the conclusion.

### Minor concern 2: inconsistent values and protocol names

*The reported scGPT plus Logistic Regression values differ between Tables 2 and 3. The protocol difference should be stated more prominently and pipelines should be named consistently.*

**Response.** The manuscript now contains one official scGPT embedding path and one set of pipeline names. The apparent numerical ambiguity has been removed. Where two summaries of the same predictions differ, the table note identifies the mathematical reason: one is pooled from all out-of-fold cells and the other is the unweighted mean of five fold-level Macro F1 values. Neither is a second fitting protocol.

### Minor concern 3: gene regulatory network terminology

*The term GRN inference may be too strong because the analysis is based on cosine similarity between gene embeddings and lacks comparison with simpler expression-based or network baselines.*

**Response.** We agree. Gene-similarity and GRN claims are outside the controlled annotation question and are not part of this manuscript. The paper is limited to annotation discrimination, probability quality, scarcity and biological error patterns.

#### Minor concern 4: reliability-bin counts and uncertainty

*The reliability diagrams do not show the number of observations or uncertainty within each confidence bin.*

**Response.** All reliability data now include bin edges, counts, mean confidence and empirical accuracy. The Supplementary Information includes large-format diagrams and the plotted CSV values. Donor-level uncertainty is used for pipeline comparisons; the manuscript avoids presenting cell-bin uncertainty as biological replication.

#### Minor concern 5: distinguish scGPT from the complete pipeline

*Throughout the text and figures scGPT should be distinguished consistently from scGPT embeddings plus downstream classifier.*

**Response.** Done. The distinction is enforced in prose, figures, legends and table labels. Calibration statements refer to, for example, “scGPT embeddings + XGBoost”, not to scGPT intrinsically.

### Additional safeguards incorporated in the final analysis

Before preparing the submission files, we conducted an additional statistical and biological audit. This produced the following safeguards:

- **Representation-specific regularization.** Logistic Regression searches seven values of  $C$  with and without feature scaling separately for every representation. Random Forest and XGBoost also have independent training-only searches.
- **No test-selected head.** The primary comparisons keep Logistic Regression, Random Forest or XGBoost fixed. A “best observed classifier” is not used for confidence intervals.
- **Compression control.** Training-only SVD-512 separates an ordinary dimensionality-reduction effect from a scGPT-specific representation effect.
- **Biological errors.** Per-class precision, recall, F1, support and full confusion matrices are reported. The main text examines lymphocyte-boundary errors in TNBC and PBMC and rare macrophage, vascular and precursor-cell errors in brain.
- **Cross-species scope.** Pig pancreas is not counted as a human benchmark. Its direct symbol matching without orthology is stated in the Methods, table note and result heading.
- **Reference quality.** Broken diacritics were repaired, dataset source publications and direct UUIDs were verified against the H5AD metadata, and the bibliography was rebuilt in LaTeX-safe form.

## Submission files

The package contains the clean manuscript, marked manuscript, Supplementary Information, this response, cover letter, separate high-resolution figures, full LaTeX source, compact result tables, grouping and checkpoint audits, executable analysis scripts and a checksum manifest. A machine-readable coverage table identifies the two unavailable PBMC SVD tree pipelines, which are left blank rather than imputed. The manuscript and Supplementary Information compile directly from the supplied source.

We thank the Editor and Reviewers again. Their comments led to a more realistic biological holdout, a fairer representation comparison and a substantially stronger calibration analysis.

Sincerely,

Anna Supernat, on behalf of all authors